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Assessment Submission Form

Student Number (If this is group work, please include the student numbers of all group participants)	Daria Karakalchev - GH1050200
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Module Code	B105B (SS0626)
Module Title	Applied Statistical Modelling
Module Tutor	WILLIAM MORRISON
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Steam Store Games

1. Business Problem

Game companies, developers and publishers must always be aware of the fact that their decisions (such as the price or even the release date) may affect not only how many customers they get, but also many other various factors. As games are a part of the entertainment industry that keeps rapidly changing, these decisions end up oftentimes critical for the success of a game. Marketing uses a lot of resources to figure out these details. Due to this, data insights on which factors associated with games can affect the commercial reach or even the player reception can be very helpful. Not only would it reduce the risk of a game becoming a failure, but also help understand the gamer community better and possibly improve the marketing strategy for future releases.

Business questions:

- Does a game's price affect the amount of people that play the game?
- Does a game's genre affect whether or not the gamers receive it positively?
- Do free to play games receive a different review ratio (positive ratings ÷ all ratings) compared to paid games?
- Is there a relationship between the amount of achievements there are in a game and how long the players will play it for?
- Is there a connection between the availability on a platform (Windows only versus other platforms) and the ownership amount of a game?

Declaration of Authorship

I declare that all material in this assessment is my own work except where there is clear acknowledgement and appropriate reference to the work of others.

I fully understand that the unacknowledged inclusion of another person's writings or ideas or works in this work may be considered plagiarism and that, should a formal investigation process confirm the allegation, I would be subject to the penalties associated with plagiarism, as per GISMA Business School, University of Applied Sciences' regulations for academic misconduct.

Signed. **Daria Karakalchev** ... Date **16.09.2026**.....

Hypotheses to be tested:

- Hypothesis 1: Price versus owners

H0: There is no significant correlation between a game's price and the amount of owners.

H1: There is a significant correlation between a game's price and the amount of owners.

- Hypothesis 2: Genre versus reception

H0: The mean review ratio does not significantly differ across genres.

H1: The mean review ratio differs significantly across genres.

- Hypothesis 3: Price versus reviews

H0: There is no significant difference in review ratios between free to play and paid games.

H1: There is a significant difference in review ratios between free to play and paid games.

- Hypothesis 4: Achievements versus play time

H0: There is no significant correlation between the amount of achievements a game has and the average playtime of gamers.

H1: There is a significant correlation between the amount of achievements a game has and the average playtime of gamers.

- Hypothesis 5: Availability on a platform versus ownership

H0: There is no significant difference in the amount of owners between Windows-only games and games that are available on other platforms.

H1: There is a significant difference in the amount of owners between Windows-only games and games that are available on other platforms.

2. Data Exploration

```
library(tidyverse)
library(psych)

steam <- read_csv("C:/Users/daria/Desktop/Final Project Applied Statistical Modelling/steam.csv")
summary(steam)
```

[illegible]

From running the functions `summary`, `glimpse` and `sapply`, it is easier to identify the data type for each of the features:

- Appid: numeric
- Name: character
- Release_date: date
- English: numeric
- Developer: character
- Publisher: character
- Platforms: character
- Required_age: numeric
- Categories: character

- Genres: character
- Steampsy_tags: character
- Achievements: numeric
- Positive_ratings: numeric
- Negative_ratings: numeric
- Average_playtime: numeric
- Median_playtime: numeric
- Owners: character
- Price: numeric

Most of the data types are exactly as they are expected to be: either character or numeric. The only one that particularly stands out is that `release_date` too was detected as date, so it requires no conversion.

The only column that is not the data type that it should be is `owners`, which is currently set as character. This is crucial, as the ‘owners’ column will be used for one of the hypotheses. Due to this, it needs to be properly converted into a numeric value, despite currently being a range string.

3. i. Data Preprocessing: Data Types

```
> unique(steam$owners)
[1] "10000000-20000000" "5000000-10000000" "2000000-5000000" "20000000-50000000" "100000000-200000000"
[6] "50000000-100000000" "20000-50000" "500000-1000000" "100000-200000" "50000-100000"
[11] "1000000-2000000" "200000-500000" "0-20000"
```

We start by checking each individual string range. We then split each range into two numbers, where the midpoint becomes the estimate that we use.

```
steam <- steam %>%
  separate(owners, into = c("owners_min", "owners_max"), sep = "-", remove = FALSE) %>%
  mutate(
    owners_min = as.numeric(owners_min),
    owners_max = as.numeric(owners_max),
    owners_estimate = (owners_min + owners_max) / 2
  )
```

The two columns are split into two and by adding an extra argument of “`remove=FALSE`”, the owner column is kept intact. The left is treated as the minimum and the right as the maximum, which are converted into numbers to find the midpoint.

```
> steam %>% select(owners, owners_min, owners_max, owners_estimate)
# A tibble: 27,075 × 4
  owners                owners_min owners_max owners_estimate
  <chr>                <dbl>      <dbl>      <dbl>
1 10000000-20000000    10000000    20000000    15000000
2 5000000-10000000     5000000     10000000     7500000
3 5000000-10000000     5000000     10000000     7500000
4 5000000-10000000     5000000     10000000     7500000
5 5000000-10000000     5000000     10000000     7500000
6 5000000-10000000     5000000     10000000     7500000
7 5000000-10000000     5000000     10000000     7500000
8 10000000-20000000    10000000    20000000    15000000
9 5000000-10000000     5000000     10000000     7500000
10 10000000-20000000    10000000    20000000    15000000
# i 27,065 more rows
# i Use `print(n = ...)` to see more rows
```

A check to see if they properly converted into numbers.

3. ii. Data Preprocessing: Missing Values

```
> colSums(is.na(steam))
  appid      name release_date  english  developer  publisher  platforms
      0         0           0         0         0         6         0
  required_age categories      genres  steamspy_tags  achievements  positive_ratings  negative_ratings
      0         0           0         0         0         0         0
  average_playtime median_playtime  owners  owners_min  owners_max      price  owners_estimate
      0         0           0         0         0         0         0
```

There are only 6 NA values, which are in the publisher column and not used for any of the hypotheses. It will be dropped later to remove the noise from the dataset.

3. ii. Data Preprocessing: Checking Duplicates

```
> sum(duplicated(steam$appid))
[1] 0
```

As this function gives 0 as an output, it means that each row contains a different game as well as an individual ID.

3. ii. Data Preprocessing: Deriving Columns

First Mutation:

```
> steam <- steam %>%
+   mutate(primary_genre = str_split(genres, ";", simplify = TRUE)[, 1])
Error in .rs.exprMutatesPackageLibrary(part) :
  argument "part" is missing, with no default
> steam %>% select(genres, primary_genre) %>% head(10)
# A tibble: 10 × 2
  genres primary_genre
  <chr>   <chr>
1 Action Action
2 Action Action
3 Action Action
4 Action Action
5 Action Action
6 Action Action
7 Action Action
8 Action Action
9 Action Action
10 Action Action
```

H2 focuses on testing the mean review and how it differs across genres, hence categorizing each game into one genre. As ANOVA cannot work with all of the genres simultaneously, although not ideal, this simplification is necessary.

To add on, although an error mirror is displayed, despite the troubleshooting and restarting the session, I found out that it was an error that Rstudio raised by itself. I could not find a solution and after checking if the code ran successfully, I found out that it did in fact create the primary_genre column, so I left it as it is.

```
> steam %>% filter(str_detect(genres, ";")) %>% select(genres, primary_genre) %>% head(10)
# A tibble: 10 × 2
  genres primary_genre
  <chr>   <chr>
1 Action;Free to Play Action
2 Action;Free to Play;Strategy Action
3 Action;Adventure Action
4 Action;Free to Play Action
5 Indie;Strategy Indie
6 Indie;Strategy Indie
7 Indie;Strategy Indie
8 Animation & Modeling;Video Production Animation & Modeling
9 Action;RPG Action
10 Action;Indie;RPG Action
```

The first 10 rows were the exact same, so there was no way to confirm whether primary_genre ran properly or not. To clearly identify the difference between the two columns, the filter function was run that focused on genres that had a semicolon, hence had multiple genres.

Second Mutation:

```
> steam <- steam %>%
+   mutate(windows_only = platforms == "windows")
> steam %>% select(platforms, windows_only) %>% head(10)
# A tibble: 10 × 2
  platforms      windows_only
  <chr>          <lgl>
1 windows;mac;linux FALSE
2 windows;mac;linux FALSE
3 windows;mac;linux FALSE
4 windows;mac;linux FALSE
5 windows;mac;linux FALSE
6 windows;mac;linux FALSE
7 windows;mac;linux FALSE
8 windows;mac;linux FALSE
9 windows;mac;linux FALSE
10 windows;mac;linux FALSE
```

This time the mutation filters out so that the “Windows_only” has a logical true or false depending on whether or not the platform is listed as only Windows. This is needed by the fifth hypothesis and confirms the availability on the specific platform, in this case Windows.

Third Mutation:

```

> steam <- steam %>%
+   mutate(total_ratings = positive_ratings + negative_ratings)
> steam <- steam %>%
+   mutate(review_ratio = positive_ratings / total_ratings)
> steam %>% select(positive_ratings, negative_ratings, total_ratings, review_ratio) %>% head(10)
# A tibble: 10 × 4
  positive_ratings negative_ratings total_ratings review_ratio
    <dbl>         <dbl>         <dbl>         <dbl>
1    124534         3339         127873         0.974
2      3318          633          3951         0.840
3      3416          398          3814         0.896
4       1273          267          1540         0.827
5       5250          288          5538         0.948
6       2758          684          3442         0.801
7      27755         1100         28855         0.962
8      12120         1439         13559         0.894
9       3822          420          4242         0.901
10     67902         2419         70321         0.966
> |

```

Both hypothesis 2 and 3 require a review ratio, which is the positive_ratings column divided by the total_ratings (so positive and negative ratings combined).

```

> sum(is.nan(steam$review_ratio))
[1] 0

```

It is important to check if any of the review_ratio rows have a missing value, which would have been the case if any of the games had zero total ratings.

—

Fourth Mutation:

```

> steam <- steam %>%
+   mutate(free_to_play = price == 0)
> |

```

```

> steam %>% select(price, free_to_play) %>% head(20)
# A tibble: 20 × 2
  price free_to_play
  <dbl> <lgl>
1  7.19 FALSE
2  3.99 FALSE
3  3.99 FALSE
4  3.99 FALSE
5  3.99 FALSE
6  3.99 FALSE
7  7.19 FALSE
8  7.19 FALSE
9  3.99 FALSE
10 7.19 FALSE
11 7.19 FALSE
12 0     TRUE
13 7.19 FALSE
14 3.99 FALSE
15 0     TRUE
16 0     TRUE
17 5.79 FALSE
18 7.19 FALSE
19 5.79 FALSE
20 0     TRUE

```

The third hypothesis requires a free-to-play column, hence it was created and the first 20 rows were checked.

3. iii. Data Preprocessing: Dropping Columns

Before dropping columns to remove noise, which columns are needed and which are not needs to be considered.

Columns to keep:

Identifiers: appid, name

H1: price, owners_estimate

H2: primary_genre, review_ratio

H3: review_ratio, free_to_play

H4: average_playtime, achievements

H5: owners_estimate, windows_only

Columns to be removed:

Release_date, English, developer, publisher, platforms, required_age, categories, genres, steamspy_tags, positive_ratings, negative_ratings, median_playtime, owners, total_ratings, owners_min, owners_max.

```
> steam <- steam %>%
+   select(appid, name, price, owners_estimate, primary_genre, review_ratio,
+   achievements, average_playtime, windows_only, free_to_play)
> glimpse(steam)
Rows: 27,075
Columns: 10
$ appid      <dbl> 10, 20, 30, 40, 50, 60, 70, 80, 130, 220, 240, 280, 300, 320, 340, 360, 380, 400, 420, 440, 500, 55...
$ name       <chr> "Counter-Strike", "Team Fortress Classic", "Day of Defeat", "Deathmatch Classic", "Half-Life: Oppos...
$ price      <dbl> 7.19, 3.99, 3.99, 3.99, 3.99, 3.99, 7.19, 7.19, 3.99, 7.19, 7.19, 0.00, 7.19, 3.99, 0.00, 0.00, 5.7...
$ owners_estimate <dbl> 1.5e+07, 7.5e+06, 7.5e+06, 7.5e+06, 7.5e+06, 7.5e+06, 7.5e+06, 1.5e+07, 7.5e+06, 1.5e+07, 1.5e+07, ...
$ primary_genre <chr> "Action", "Action", "Action", "Action", "Action", "Action", "Action", "Action", "Action", "Action", ...
$ review_ratio <dbl> 0.9738882, 0.8397874, 0.8956476, 0.8266234, 0.9479957, 0.8012783, 0.9618784, 0.8938712, 0.9009901, ...
$ achievements <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 33, 147, 0, 54, 0, 0, 0, 13, 15, 22, 520, 73, 70, 0, 51, 66, 167, 0, 44, ...
$ average_playtime <dbl> 17612, 277, 187, 258, 624, 175, 1300, 427, 361, 691, 6842, 190, 1356, 311, 46, 102, 281, 288, 354, ...
$ windows_only <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, F...
$ free_to_play <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, TRUE, FALSE, FALSE, TR...
```

By using the select function, the columns that were not used anywhere were dropped from the dataset.

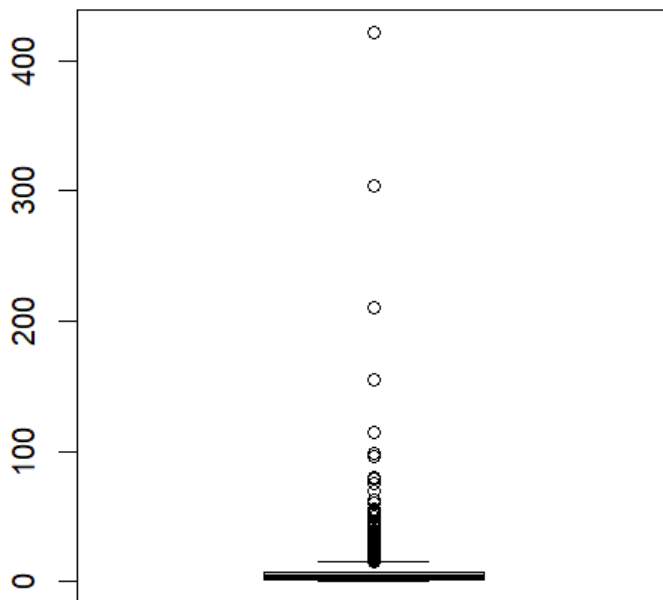
3. iv. Data Preprocessing: Checking Outliers

Outliers will be checked for numerical columns, which are currently as follows:
Price, owners_estimate, achievements, average_playtime, review_ratio.

- First Outlier Check: Price

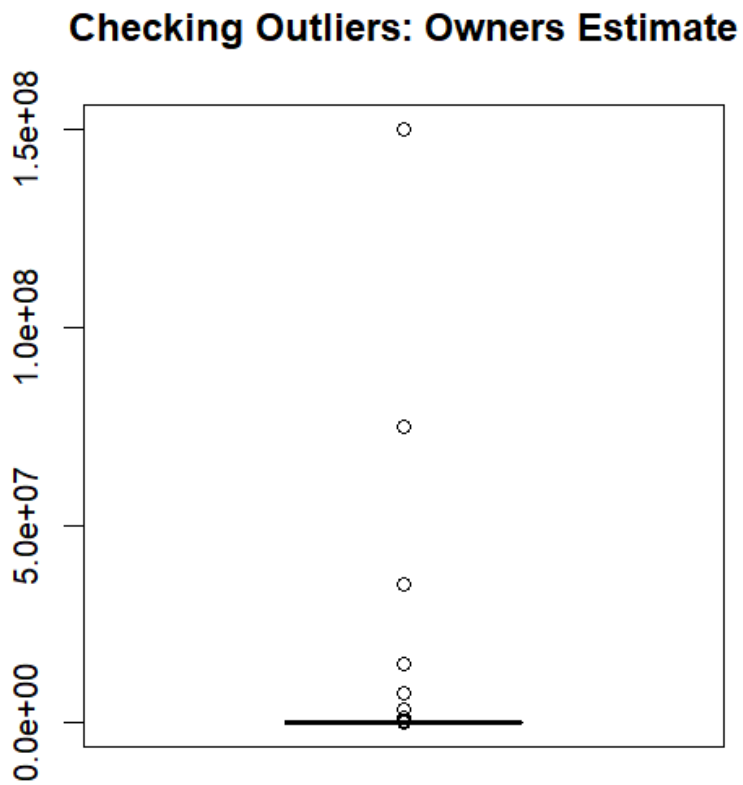
```
> summary(steam$price)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.000   1.690   3.990   6.078   7.190 421.990
> boxplot(steam$price, main = "Checking Outliers: Price")
~ |
```

Checking Outliers: Price



- Second Outlier Check: Owners_estimate

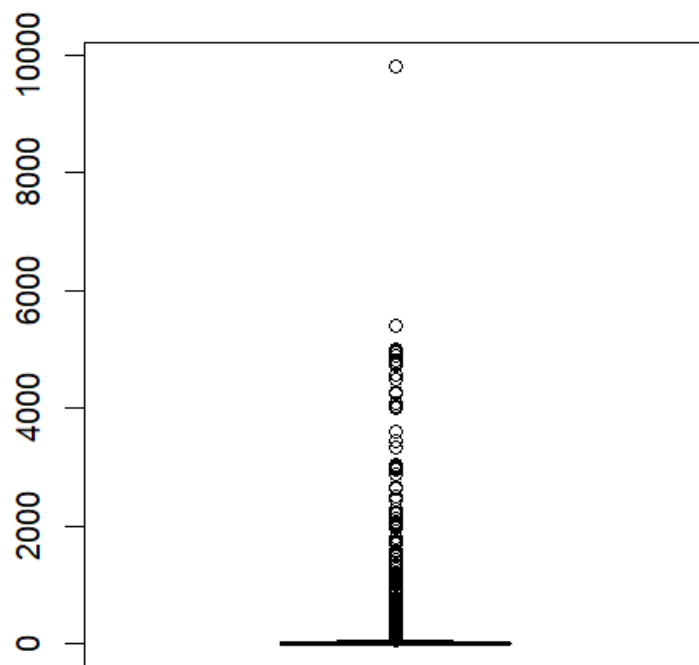
```
> summary(steam$owners_estimate)
   Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
 10000   10000   10000  134090   35000 150000000
> boxplot(steam$owners_estimate, main = "Checking Outliers: Owners Estimate")
```



-
- Third Outlier Check: Achievements

```
> summary(steam$achievements)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.00   0.00    7.00   45.25   23.00 9821.00
> boxplot(steam$achievements, main = "Checking Outliers: Achievements")
\ |
```

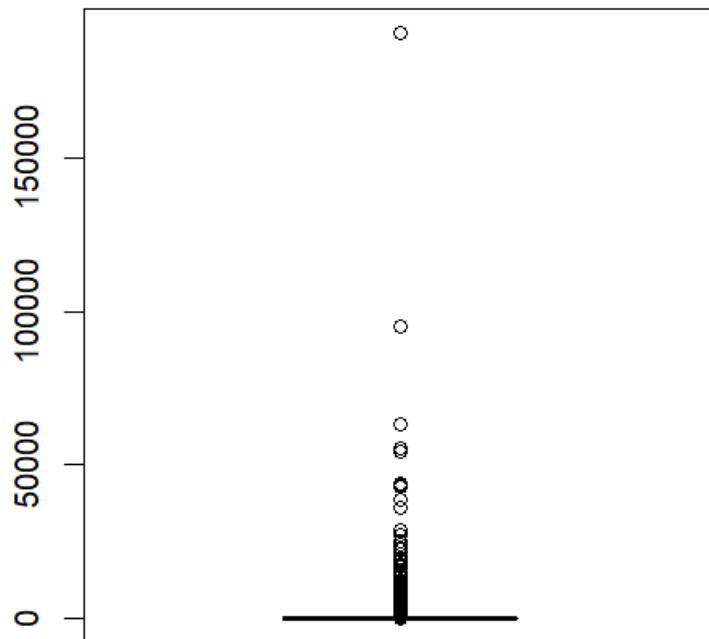
Checking Outliers: Achievements



- Fourth Outlier Check: Average_playtime

```
> summary(steam$average_playtime)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0    0.0    0.0   149.8    0.0 190625.0
> boxplot(steam$average_playtime, main = "Checking Outliers: Average Playtime")
```

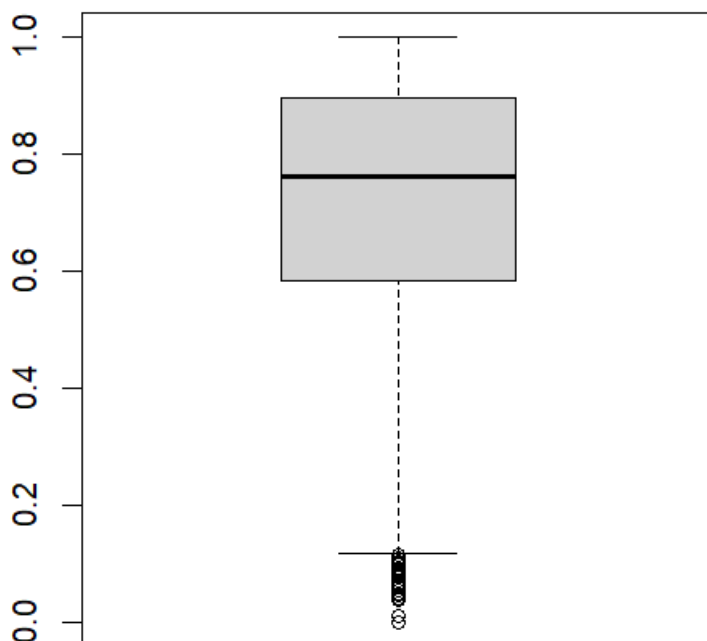
Checking Outliers: Average Playtime



- Fifth Outlier Check: Review_ratio

```
> summary(steam$review_ratio)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.5833 0.7603 0.7145 0.8939 1.0000
> boxplot(steam$review_ratio, main = "Checking Outliers: Review Ratio")
\ |
```

Checking Outliers: Review Ratio



3. v. Data Preprocessing: Sampling Decision

As found through the glimpse function, this dataset contains 27075 rows. This is considered quite big for a dataset, but despite this, R is handling all of the data with no issues. Removing any of the data would significantly reduce the effectiveness of any of the conclusions made in the future, which is why I decided to use the entire dataset.

4. i. Exploratory Data Analysis: Descriptive Statistics

```
> describe(steam)
  vars      n    mean      sd   median trimmed    mad   min      max      range  skew kurtosis
appid      1 27075 596203.51 250894.17 599070.00 602523.30 294770.53    10 1.06946e+06 1.06945e+06 -0.22   -0.65
name*      2 27075 13514.57   7804.06 13514.00 13514.17 10020.89     1 2.70330e+04 2.70320e+04  0.00   -1.20
price      3 27075    6.08     7.87    3.99    4.85     4.45     0 4.21990e+02 4.21990e+02 11.73   420.71
owners_estimate 4 27075 134090.49 1328088.58 10000.00 25196.44     0.00 10000 1.50000e+08 1.49990e+08 69.81  6798.37
primary_genre* 5 27075    6.00     6.01    3.00    4.64     1.48     1 2.60000e+01 2.50000e+01  1.70    1.96
review_ratio   6 27075    0.71     0.23    0.76    0.74     0.22     0 1.00000e+00 1.00000e+00 -1.01    0.73
achievements   7 27075   45.25    352.67    7.00   10.89    10.38     0 9.82100e+03 9.82100e+03 13.43  191.14
average_playtime 8 27075  149.80   1827.04    0.00   17.02     0.00     0 1.90625e+05 1.90625e+05 59.52  5081.15
windows_only   9 27075    NaN      NA      NA      NA      NA    Inf    -Inf    -Inf     NA     NA
free_to_play  10 27075    NaN      NA      NA      NA      NA    Inf    -Inf    -Inf     NA     NA
  se
appid      1524.78
name*       47.43
price        0.05
owners_estimate 8071.29
primary_genre*  0.04
review_ratio    0.00
achievements    2.14
average_playtime 11.10
windows_only    NA
free_to_play    NA
```

Both `windows_only` and `free_to_play` are logical columns, so the `describe` function cannot compute the statistics for them, instead a different function is used.

```
> table(steam$windows_only)

FALSE  TRUE
 8677 18398

> table(steam$free_to_play)

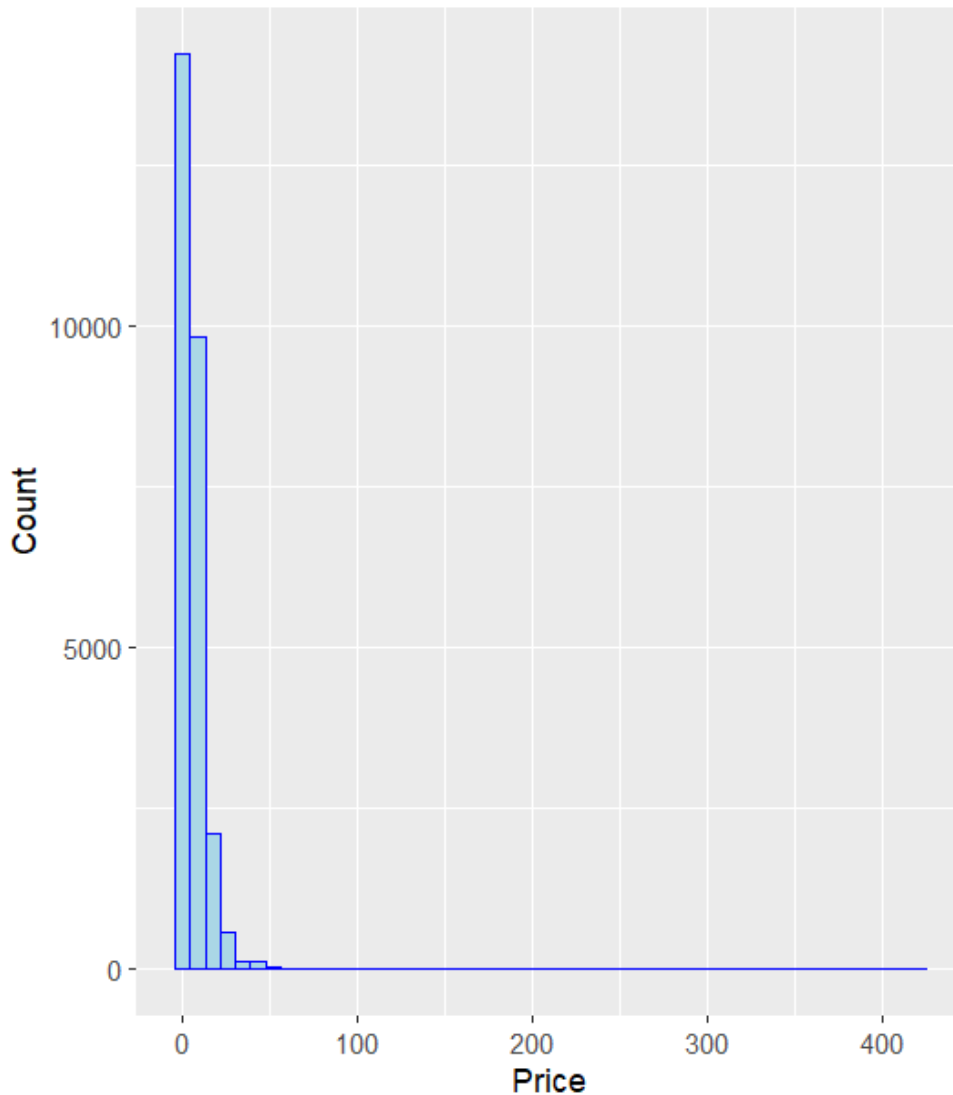
FALSE  TRUE
24515  2560
```

As the tidyverse library contains the package `ggplot`, we can start turning each column into a histogram to analyse better.

4. i. Exploratory Data Analysis: Numerical Data

- Price:

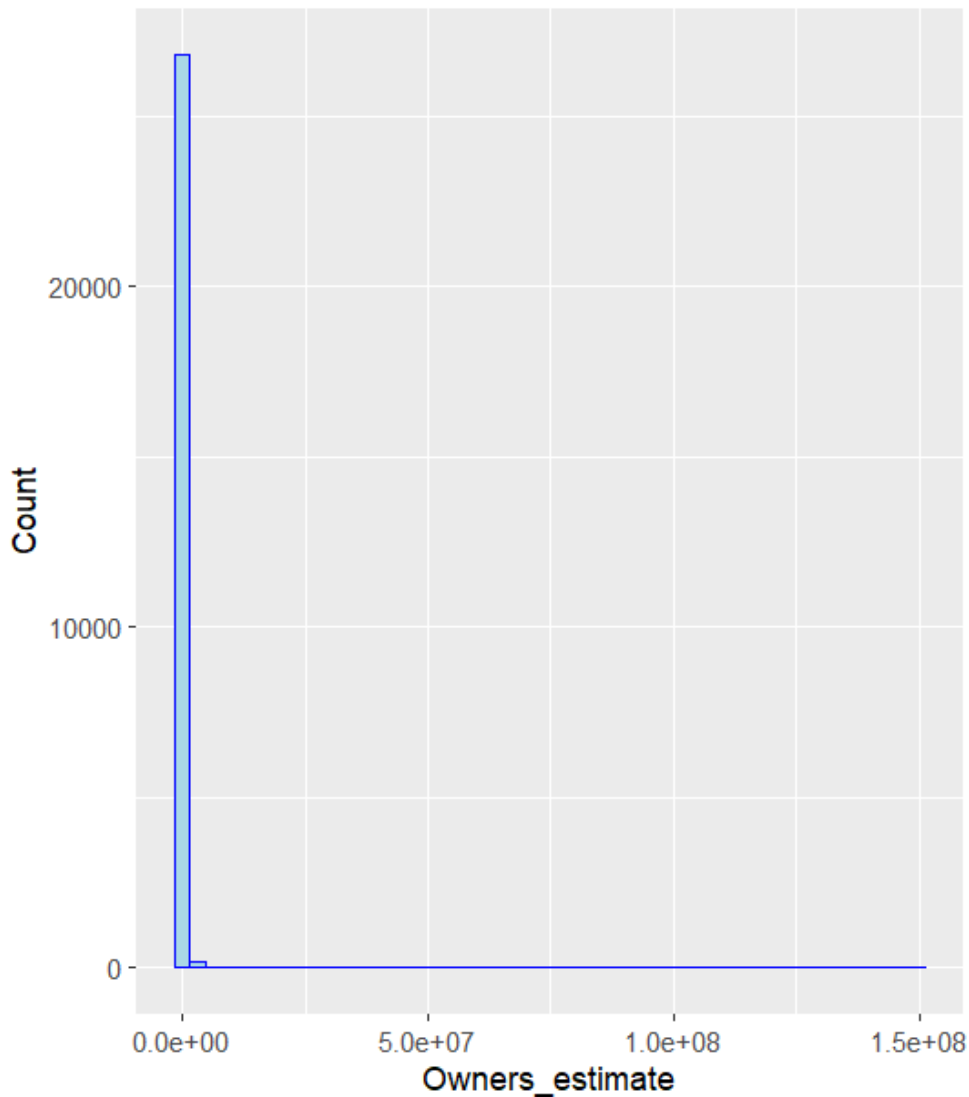
```
ggplot(steam, aes(x = price)) +
  geom_histogram(bins = 50, fill = "lightblue", color = "blue") +
  labs(x = "Price", y = "Count")
```



Above, it can be seen that this is a right skewed distribution, with most of the data being in the 0-100 range. It also contains a long tail of outliers, that go above the 400 dollars. The outliers are likely correlated with real pricing decisions, which are correlated to the first hypothesis, hence why they will be kept.

- Owners_estimate:

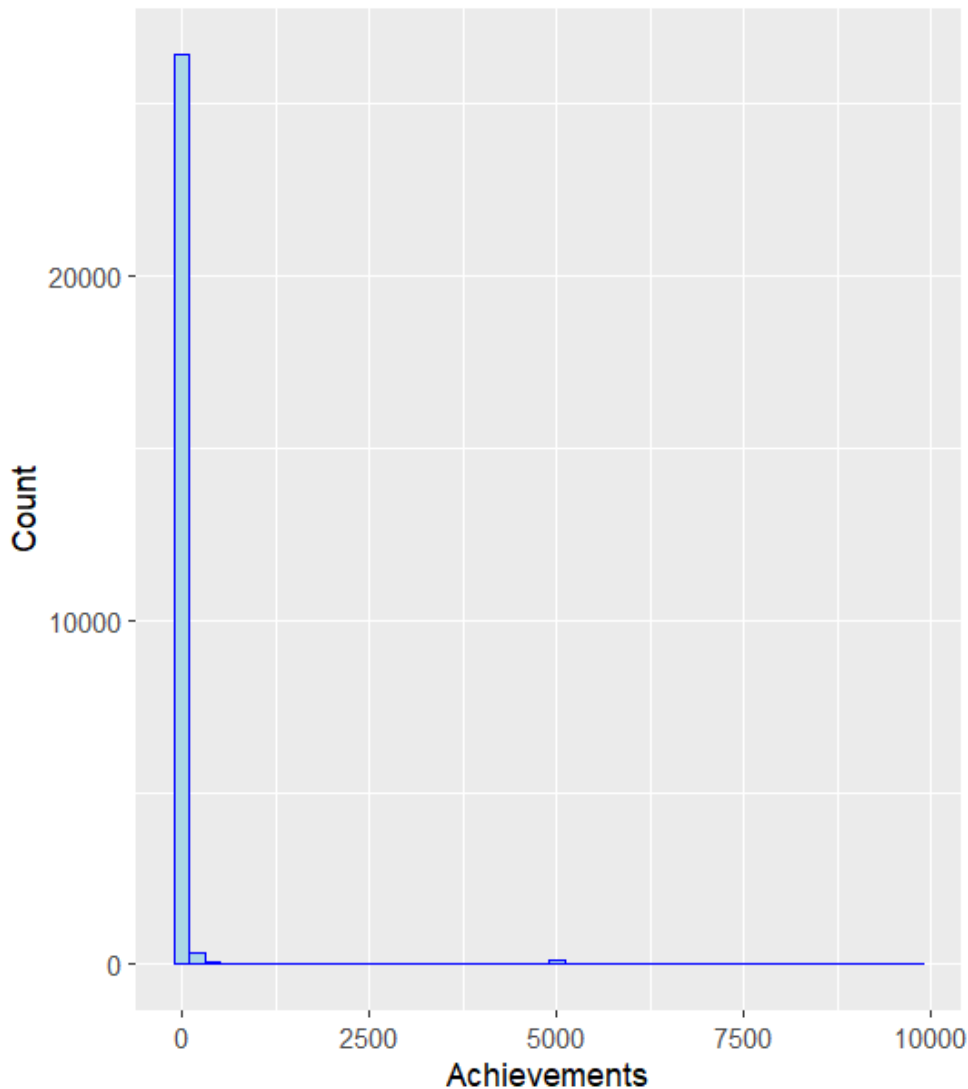
```
ggplot(steam, aes(x = owners_estimate)) +  
  geom_histogram(bins = 50, fill = "lightblue", color = "blue") +  
  labs(x = "Owners_estimate", y = "Count")
```



In the figure above, once again a right skewed distribution is observed. This time, however, the skewness is even more extreme (skewness = 69.81) with most of the games lying on the left side of the histogram. As this data will be used for hypotheses 1 and 5, it is important to keep all values to prevent bias, although a transformation will be necessary.

- Achievements:

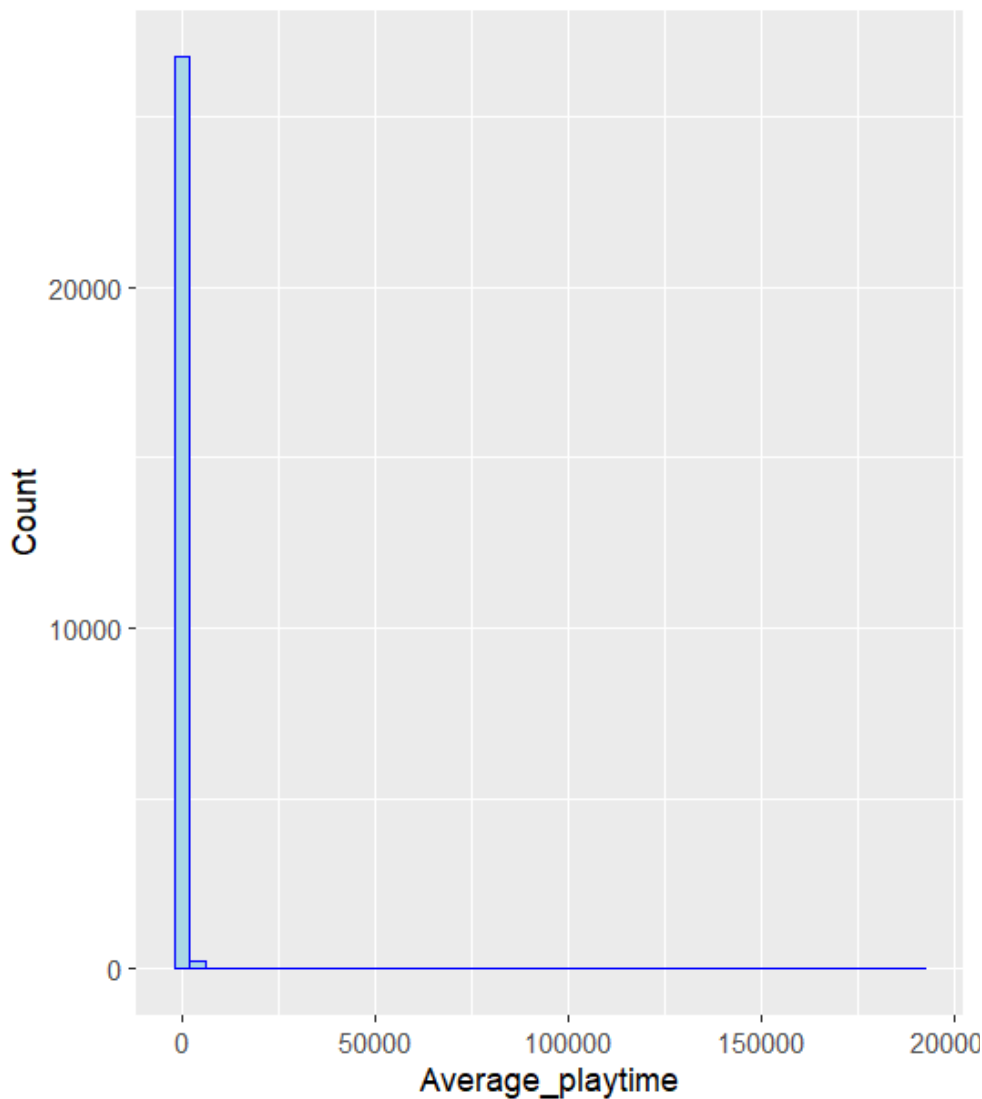
```
ggplot(steam, aes(x = achievements)) +  
  geom_histogram(bins = 50, fill = "lightblue", color = "blue") +  
  labs(x = "Achievements", y = "Count")
```



Compared to the previous two figures, although this is also a right skewed distribution, the skewness is less (skewness = 13.43). Adding more achievements to a game does not necessarily equate to a game being more successful (median = 7, shows this). Despite this, as the datapoint around 10000 represents a real entry in the dataset, it will not be removed. It should, however, be noted that it may weaken the analysis on hypothesis 4.

- Average_playtime:

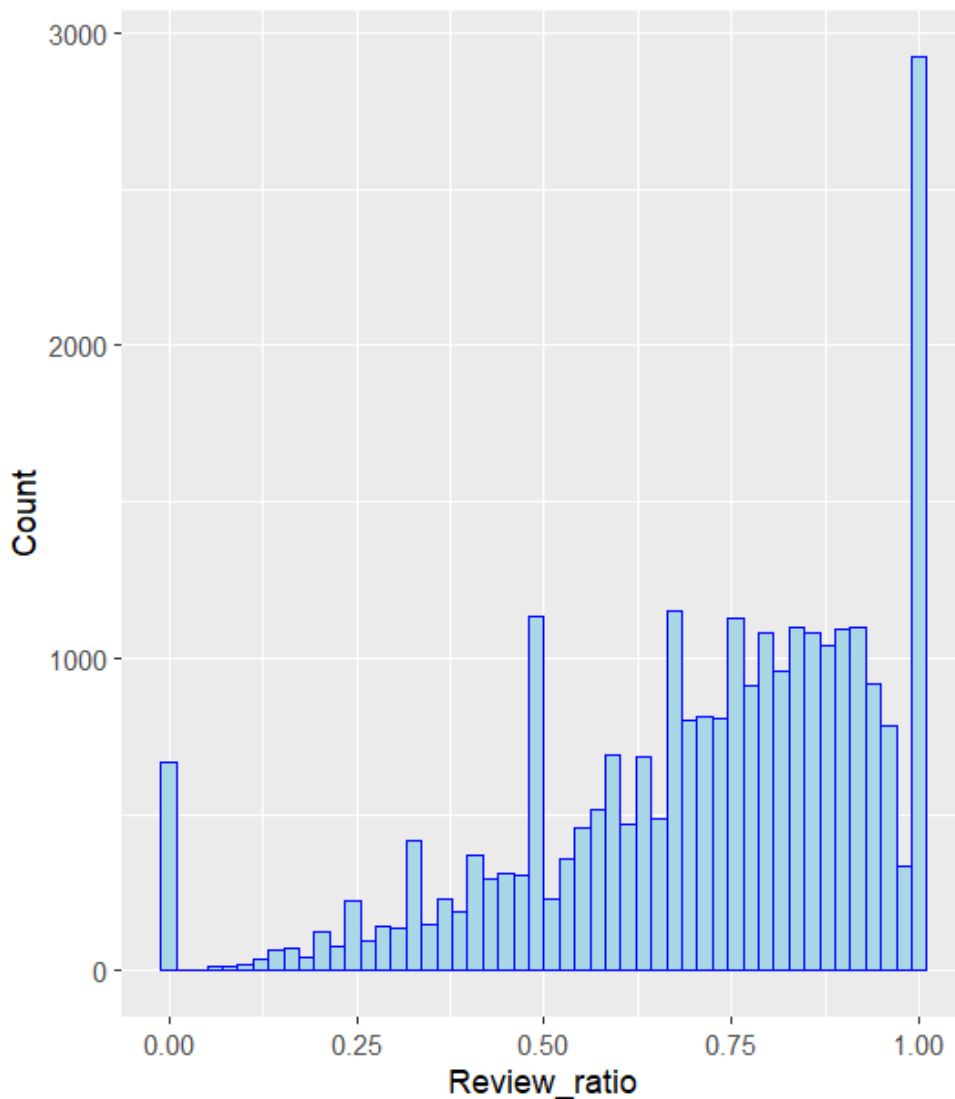
```
ggplot(steam, aes(x = average_playtime)) +  
  geom_histogram(bins = 50, fill = "lightblue", color = "blue") +  
  labs(x = "Average_playtime", y = "Count")
```



This is a heavily right skewed distribution (skewness = 59.52), with a few outliers. It is very important to note that there is no real way of being able to tell whether or not this playtime was through active engagement with the game. That being the case, once again all of the entries from the dataset were kept as could affect the evaluation on hypothesis 4. To add on the median is 0 minutes, showing that a lot of the dataset has games where there are few to none active players.

- Review_ratio:

```
ggplot(steam, aes(x = review_ratio)) +
  geom_histogram(bins = 50, fill = "lightblue", color = "blue") +
  labs(x = "Review_ratio", y = "Count")
```

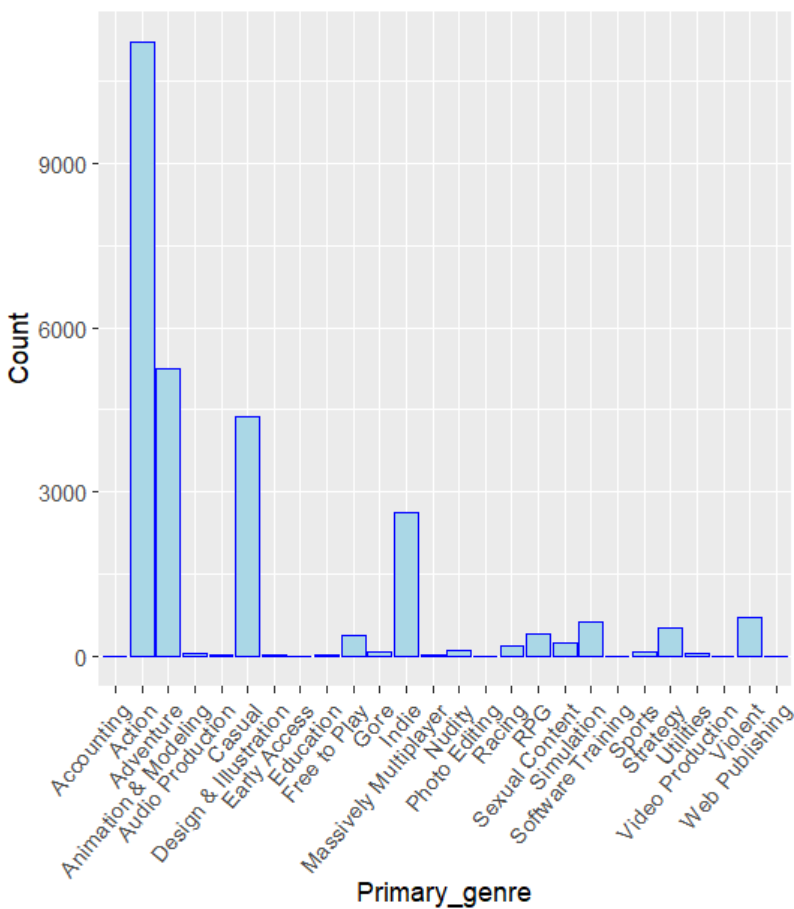


The figure above is bounded between 0 and 1 and shows a left skewed distribution. There were a few outliers that most likely represent real entries, for which they were kept as they are crucial for the observation in hypotheses 2 and 3. Additionally, as the median is 0.76, most games cluster around that range. It is also worth to note that there are quite a few sharp spikes in the distribution, which most likely represent games that had small sample sizes/not a lot of ratings, which caused these excessive spikes to form.

4. i. Exploratory Data Analysis: Categorical Data

- Primary_genre:

```
ggplot(steam, aes(x = primary_genre)) +  
  geom_bar(fill = "lightblue", color = "blue") +  
  labs(x = "Primary_genre", y = "Count") +  
  theme(axis.text.x = element_text(angle = 50, hjust = 1))
```



During this, I remembered the description of the dataset said “Unreleased titles were removed as well as many non-games like software, though some may have slipped through.” (Nik Davis, 2019).

```
> sort(table(steam$primary_genre), decreasing = TRUE)
```

Action	Adventure	Casual	Indie	Violent
11212	5256	4373	2623	709
Simulation	Strategy	RPG	Free to Play	Sexual Content
631	532	407	395	245
Racing	Nudity	Sports	Gore	Animation & Modeling
197	112	83	81	66
Utilities	Design & Illustration	Audio Production	Massively Multiplayer	Education
60	28	16	16	14
Early Access	Video Production	Accounting	Photo Editing	Software Training
9	5	2	1	1
Web Publishing				
1				

I checked to see how much this affected my previous data exploration and it was only 219 rows out of 27,075, which was around 0.8% of the entire dataset. Unfortunately affected rows were not caught earlier but considering the fact that this type of percent is not enough to affect skewness, outliers or any other values, they will be removed at this stage and not affect the hypothesis.

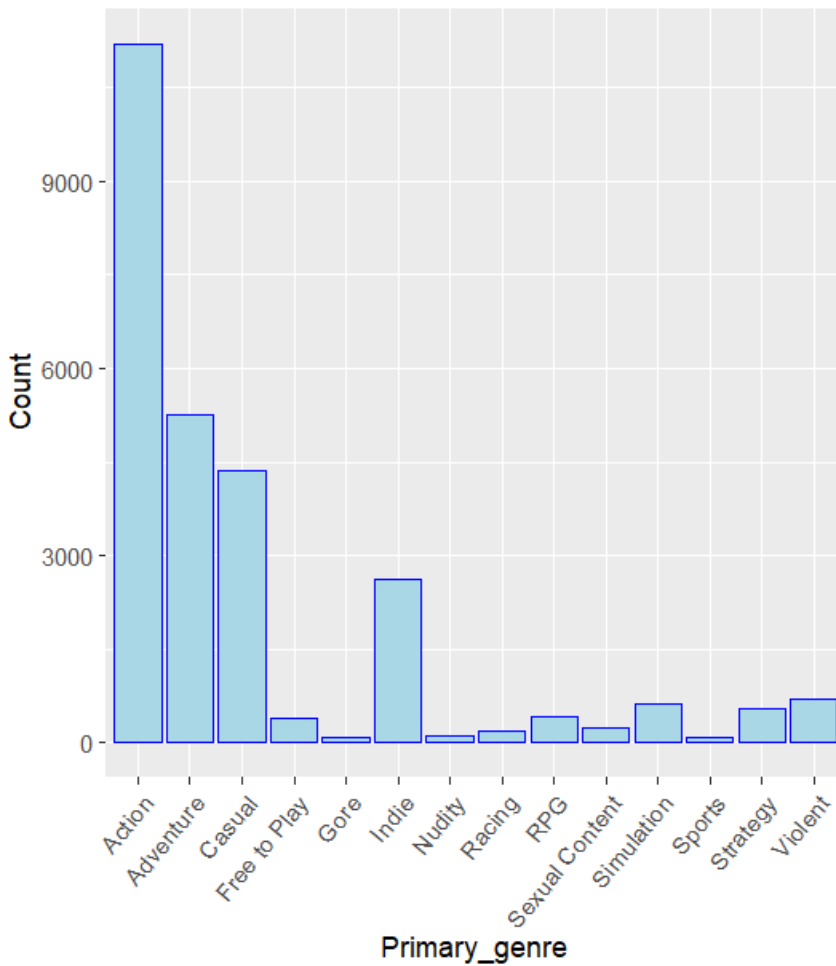
```
> genre_counts <- steam %>% count(primary_genre)
> genres_to_keep <- genre_counts %>% filter(n >= 67) %>% pull(primary_genre)
> steam <- steam %>% filter(primary_genre %in% genres_to_keep)
> sort(table(steam$primary_genre), decreasing = TRUE)
```

Action	Adventure	Casual	Indie	Violent	Simulation	Strategy	RPG
11212	5256	4373	2623	709	631	532	407
Free to Play	Sexual Content	Racing	Nudity	Sports	Gore		
395	245	197	112	83	81		

```
> nrow(steam)
[1] 26856
```

The new histogram:

```
ggplot(steam, aes(x = primary_genre)) +
  geom_bar(fill = "lightblue", color = "blue") +
  labs(x = "Primary_genre", y = "Count") +
  theme(axis.text.x = element_text(angle = 50, hjust = 1))
```

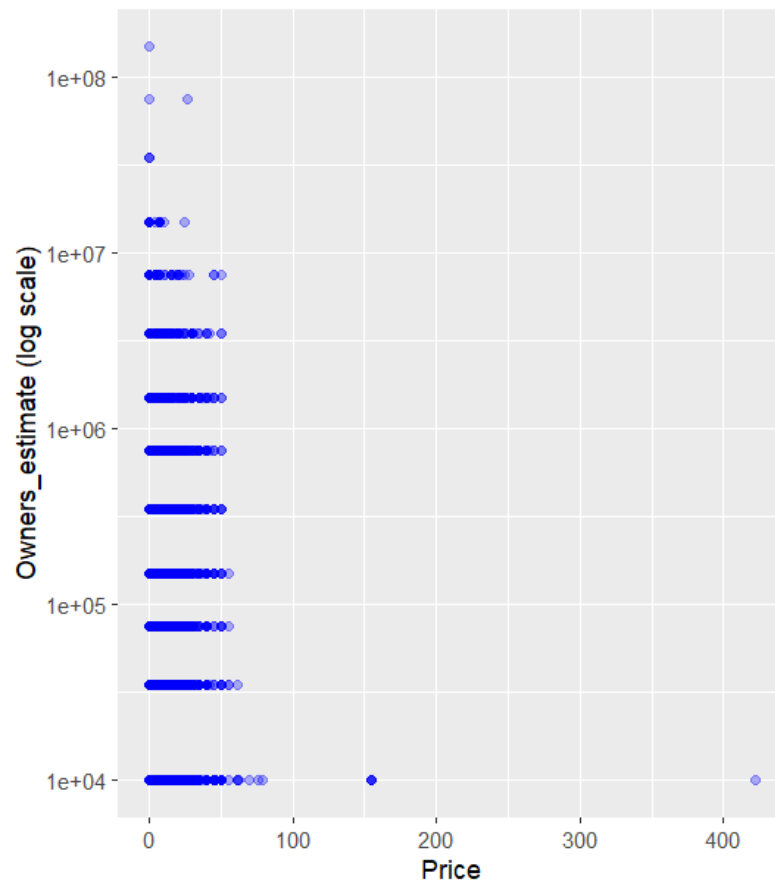


The Action column has the highest grossing value, which will likely lead to statistically reliable estimates, whilst the sparser genres will have more uncertainty.

4. ii. Exploratory Data Analysis: Scatterplots and boxplots

- Price vs owners (H1):

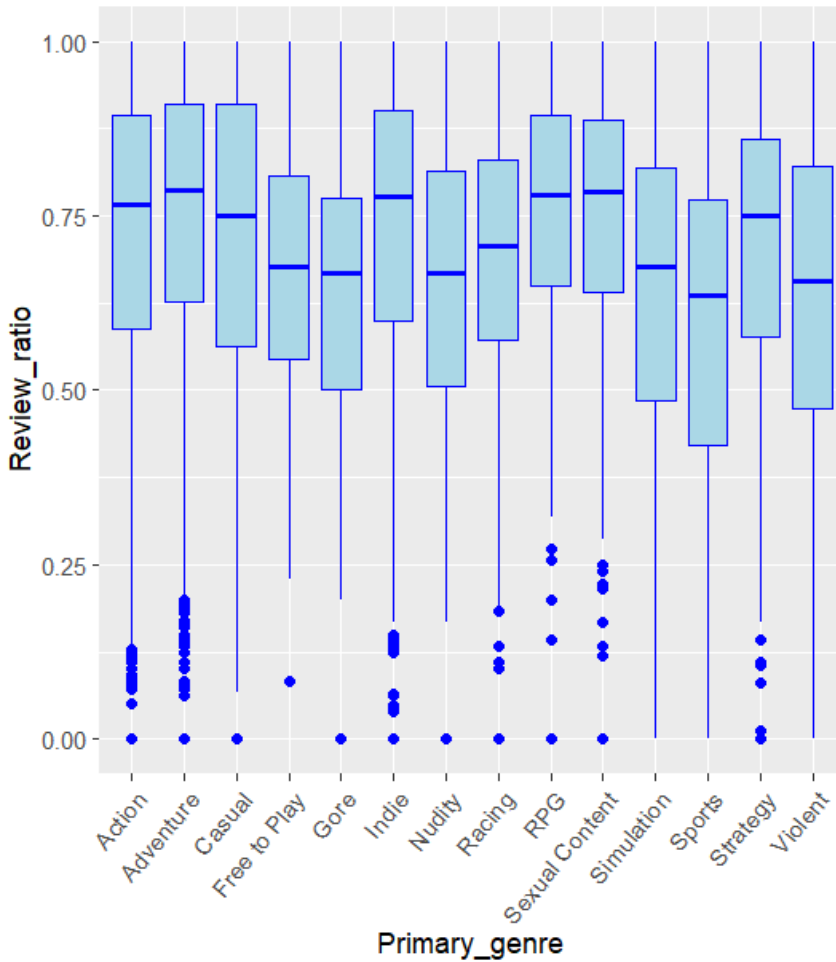
```
ggplot(steam, aes(x = price, y = owners_estimate)) +
  geom_point(alpha = 0.3, color = "blue") +
  scale_y_log10() +
  labs(x = "Price", y = "Owners_estimate (log scale)")
```



Log scale, which compresses the high values and stretches the low values, was used to be able to visualise the scatterplot better. From this, it can be seen that there is no clear trend and the points are quite evenly scattered around the plot. The points are gathered around into horizontal rows due to the fact that owners_estimate uses a midpoint of the values.

-
- Review_ratio vs primary_genre (H2)

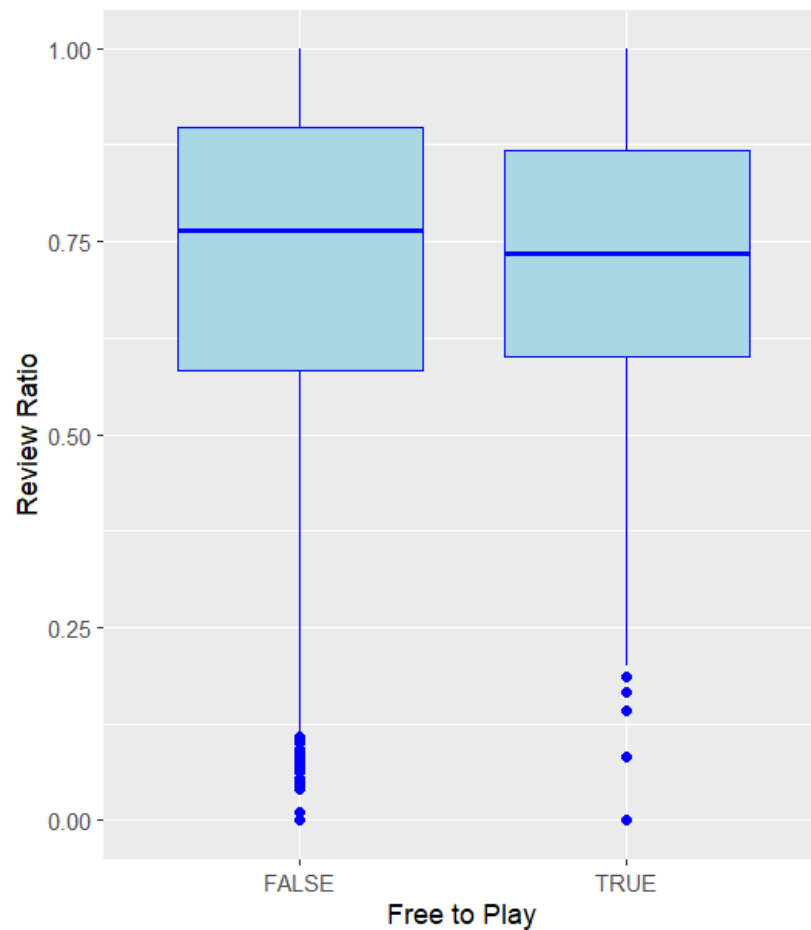
```
ggplot(steam, aes(x = primary_genre, y = review_ratio)) +
  geom_boxplot(fill = "lightblue", color = "blue") +
  labs(x = "Primary_genre", y = "Review_ratio") +
  theme(axis.text.x = element_text(angle = 50, hjust = 1))
```



There is a distinct, yet not extreme, variation in the medians across the genres, but are all in the 0.6 to 0.8 range. Sports has the widest box that shows the highest spread. Most of the genres show a few outliers towards the bottom.

- Review_ratio vs free_to_play (H3)

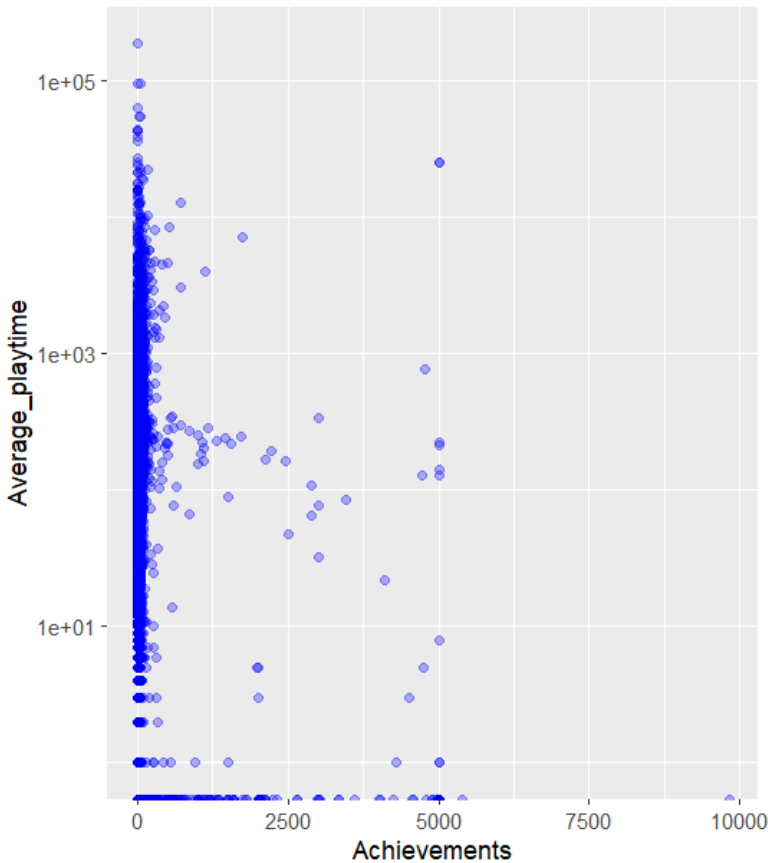
```
ggplot(steam, aes(x = free_to_play, y = review_ratio)) +
  geom_boxplot(fill = "lightblue", color = "blue") +
  labs(x = "Free_to_play", y = "Review_ratio")
```



Both the free to play and paid games are quite similar, despite the obvious difference of the paid games having a higher median.

- Achievements vs average_playtime (H4)

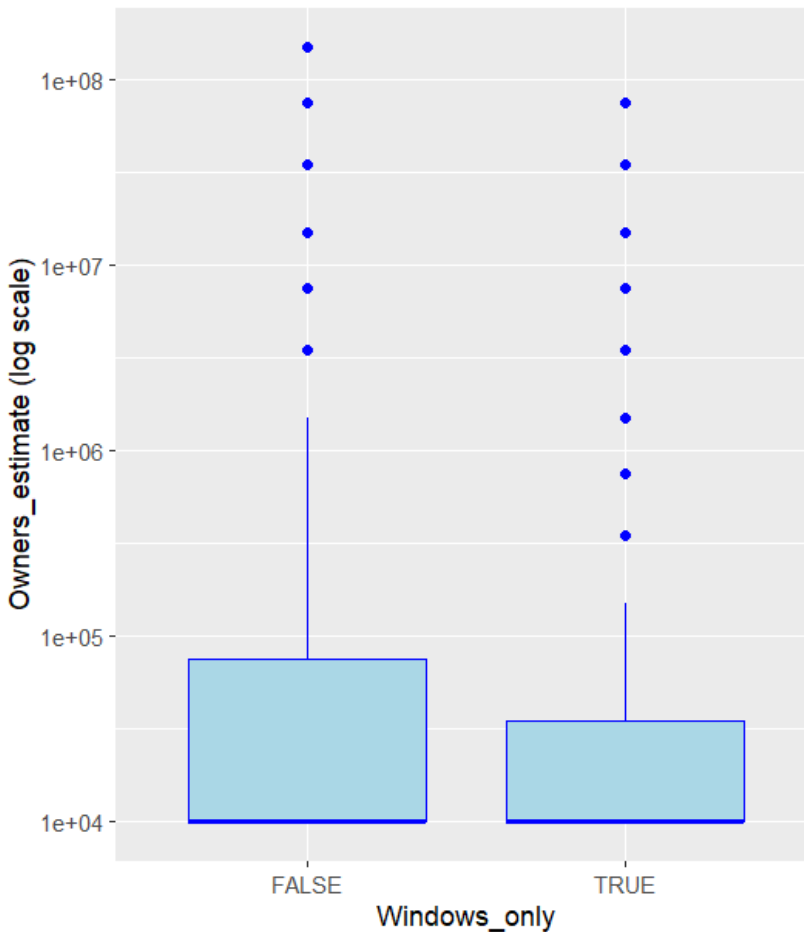
```
ggplot(steam, aes(x = achievements, y = average_playtime)) +
  geom_point(alpha = 0.3, color = "blue") +
  scale_y_log10() +
  labs(x = "Achievements", y = "Average_playtime")
```



Same here, there was a y log scale, which allowed the observation that there are no obvious positive relationships. Rather than anything, I noticed that if a game had a small amount of achievements, the playtime was higher for those games.

-
- Owners_estimate vs windows_only (H5)

```
ggplot(steam, aes(x = windows_only, y = owners_estimate)) +
  geom_boxplot(fill = "lightblue", color = "blue") +
  scale_y_log10() +
  labs(x = "windows_only", y = "Owners_estimate (log scale)")
```



Both of the boxes start from the minimum value, but the False box has a taller box showing that it is more common to have cross platform games, rather than only Windows. However as correlation is not necessarily causation, it is impossible to say whether or not having this kind of support is directly related to there being more owners.

4. iii. Exploratory Data Analysis: Correlation Matrix

```
> numeric_vars <- steam %>% select(price, owners_estimate, achievements, average_playtime, review_ratio)
> cor(numeric_vars)
```

	price	owners_estimate	achievements	average_playtime	review_ratio
price	1.000000000	0.03477788	-0.007319698	0.05270971	0.07887059
owners_estimate	0.034777882	1.00000000	0.003369950	0.17424747	0.03490466
achievements	-0.007319698	0.00336995	1.00000000	0.01434945	-0.02254697
average_playtime	0.052709712	0.17424747	0.014349450	1.00000000	0.02024154
review_ratio	0.078870590	0.03490466	-0.022546971	0.02024154	1.00000000

```
> library(corrplot)
> corrplot(cor(numeric_vars), method = "color", addCoef.col = "black", type = "upper", diag = FALSE)
```



All of the correlations in this matrix are quite weak, with the weakest being the one between achievements vs average_playtime, which is 0.01. Although it is still possible that for both Hypotheses 1 and 4, it might turn out to be statistically significant, for now it is just a possibility.

5. Selecting Inferential Tests and Validating Assumptions

- Simple Linear Regression:

H1 (price vs owners_estimate) and H4 (achievements vs average_playtime) have continuous variables, so the t-test will use the slope coefficient to identify the significance.

- One Way ANOVA:

H2 (review_ratio across primary_genre) has exactly one continuous variable, which will be compared with more than two independent groups.

- Independent Samples t-test:

H3 (review_ratio by free_to_play) and H5 (owners_estimate by windows_only) have a single continuous variable that will be compared with only two independent groups.

Bonus: If the skewness causes an issue in any of the hypotheses, hence preventing us from determining the significance, a log-scale will be used for a transformation.

6. Applying Statistical Tests

1. Hypothesis 1 (price vs owners_estimate):

```
> #Hypothesis 1
> h1_model <- lm(owners_estimate ~ price, data = steam)
> summary(h1_model)
```

Call:
lm(formula = owners_estimate ~ price, data = steam)

Residuals:

Min	1Q	Median	3Q	Max
-2785049	-125569	-99665	-85977	149904023

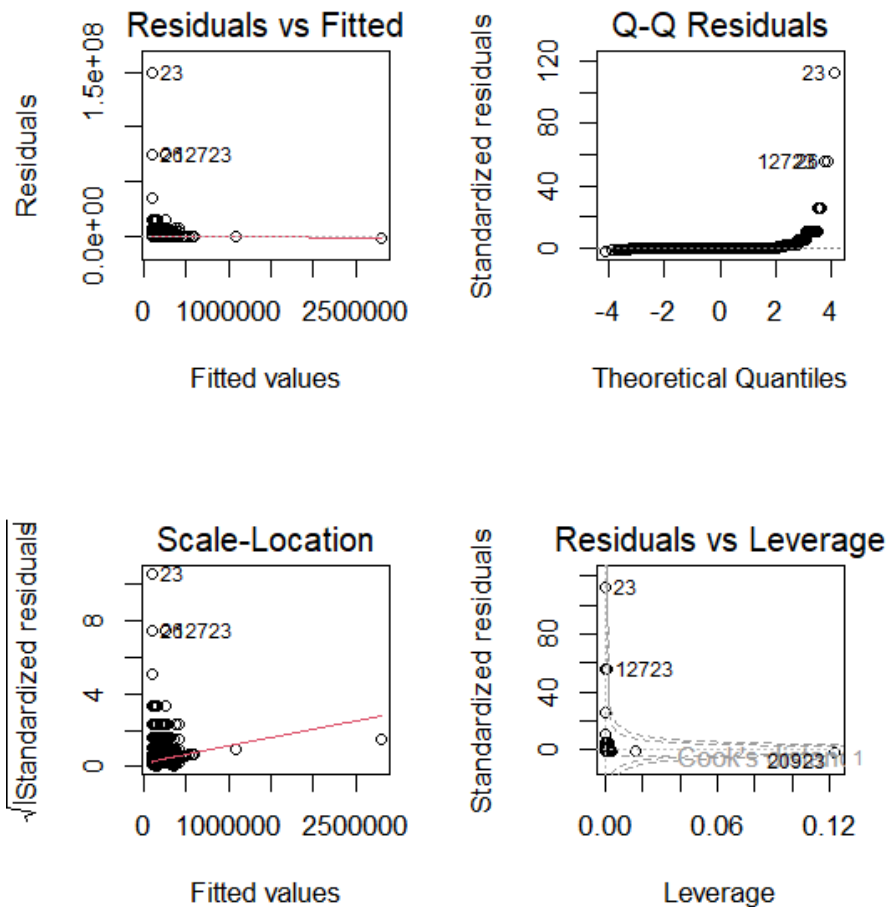
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	95977	10549	9.098	< 2e-16 ***
price	6396	1122	5.703	1.19e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1332000 on 26854 degrees of freedom
Multiple R-squared: 0.00121, Adjusted R-squared: 0.001172
F-statistic: 32.52 on 1 and 26854 DF, p-value: 1.193e-08

```
par(mfrow = c(2, 2))
plot(h1_model)
par(mfrow = c(1, 1))
```



Shapiro-Wilk normality test (it has a 5000 row limit, so we sample):

```
> set.seed(42)
> shapiro.test(sample(resid(h1_model), 5000))
```

Shapiro-Wilk normality test

```
data: sample(resid(h1_model), 5000)
W = 0.066038, p-value < 2.2e-16
```

```
> steam <- steam %>%
+   mutate(log_owners_estimate = log(owners_estimate))
> h1_model_log <- lm(log_owners_estimate ~ price, data = steam)
> summary(h1_model_log)
```

Call:

```
lm(formula = log_owners_estimate ~ price, data = steam)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-11.3126	-0.7593	-0.6681	0.5340	8.9982

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.827986	0.010553	931.30	<2e-16 ***
price	0.025344	0.001122	22.59	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.332 on 26854 degrees of freedom

Multiple R-squared: 0.01864, Adjusted R-squared: 0.01861

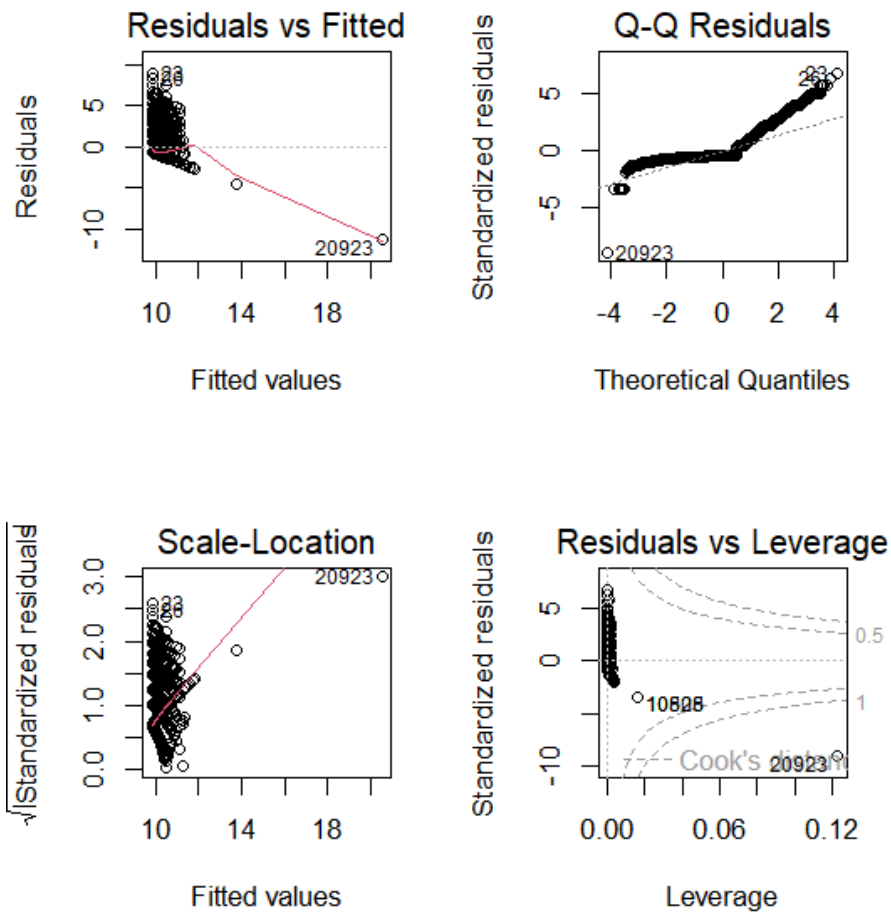
F-statistic: 510.2 on 1 and 26854 DF, p-value: < 2.2e-16

```
> par(mfrow = c(2, 2))
> plot(h1_model_log)
> par(mfrow = c(1, 1))
> set.seed(42)
> shapiro.test(sample(resid(h1_model_log), 5000))
```

Shapiro-Wilk normality test

data: sample(resid(h1_model_log), 5000)

W = 0.71256, p-value < 2.2e-16



```
> set.seed(42)
> shapiro.test(sample(resid(h1_model_log), 5000))
```

Shapiro-wilk normality test

```
data: sample(resid(h1_model_log), 5000)
W = 0.71256, p-value < 2.2e-16
```

```

> summary(h1_model_log)

Call:
lm(formula = log_owners_estimate ~ price, data = steam)

Residuals:
    Min       1Q   Median       3Q      Max
-11.3126  -0.7593  -0.6681   0.5340   8.9982

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.827986   0.010553  931.30  <2e-16 ***
price         0.025344   0.001122   22.59  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.332 on 26854 degrees of freedom
Multiple R-squared:  0.01864,    Adjusted R-squared:  0.01861
F-statistic: 510.2 on 1 and 26854 DF,  p-value: < 2.2e-16

```

Interpretation:

It can be seen that the p value is a lot less than 0.05, meaning that the relationship between price and owners_estimate is statistically significant and positive. Despite this, the r squared value is a mere 0.00121, so the variance for owners_estimate is close to 0.1%. Moving on, the assumption checks revealed clear violation of normality and very obvious clustering, due to the extreme skewness, so it had to be log transformed.

With this newly transformed model, the p value was still under 0.05, hence the relationship remained statistically significant. Additionally, the r squared value increased to 0.018, which is still under the 2% of the variance in owners_estimate. Fortunately the transformation was very useful as it improved the earlier clustering issue and allowed for more spread out data points. Despite this, the Shapiro-Wilk test still indicated non-normality, most likely due to the large sample size, which is expected. Lastly, due to the statistically significant relationship found through simple linear regression, H0 is rejected and we **accept H1**.

2. Hypothesis 2 (review_ratio across primary_genre):

```

> #Hypothesis 2
> h2_model <- aov(review_ratio ~ primary_genre, data = steam)
> summary(h2_model)
              Df Sum Sq Mean Sq F value Pr(>F)
primary_genre   13   17.5    1.346   24.93 <2e-16 ***
Residuals 26842 1449.3    0.054
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> |

```

```

> set.seed(42)
> shapiro.test(sample(resid(h2_model), 5000))

```

Shapiro-Wilk normality test

```

data: sample(resid(h2_model), 5000)
W = 0.9243, p-value < 2.2e-16

```

```

> leveneTest(review_ratio ~ primary_genre, data = steam)
Levene's Test for Homogeneity of Variance (center = median)
              Df F value    Pr(>F)
group         13  11.641 < 2.2e-16 ***
 26842

```

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

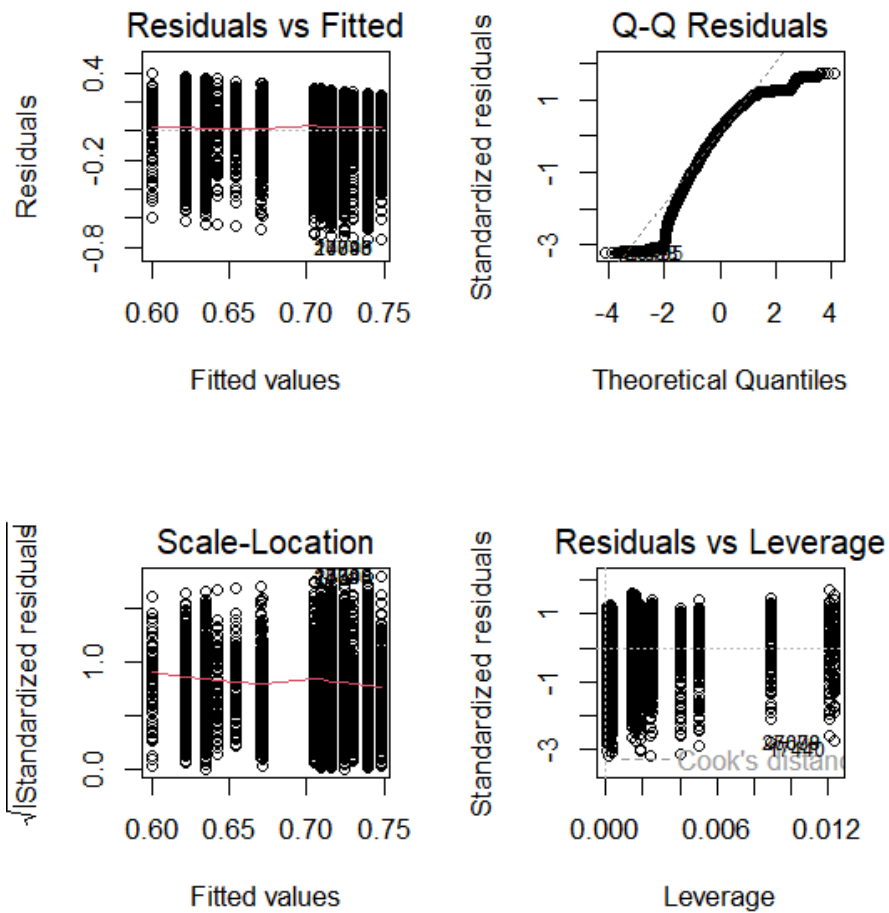
Warning message:

In leveneTest.default(y = y, group = group, ...) : group coerced to factor.

```

par(mfrow = c(2, 2))
plot(h2_model)
par(mfrow = c(1, 1))
|

```



```
> kruskal.test(review_ratio ~ primary_genre, data = steam)
```

Kruskal-Wallis rank sum test

data: review_ratio by primary_genre

Kruskal-Wallis chi-squared = 374.14, df = 13, p-value < 2.2e-16

```
> pairwise.wilcox.test(steam$review_ratio, steam$primary_genre, p.adjust.method = "bonferroni")

Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data: steam$review_ratio and steam$primary_genre
```

	Action	Adventure	Casual	Free to Play	Gore	Indie	Nudity	Racing	RPG	Sexual Content	Simulation
Adventure	1.6e-07	-	-	-	-	-	-	-	-	-	-
Casual	1.00000	0.00013	-	-	-	-	-	-	-	-	-
Free to Play	1.9e-09	< 2e-16	3.2e-07	-	-	-	-	-	-	-	-
Gore	0.02427	0.00037	0.08525	1.00000	-	-	-	-	-	-	-
Indie	1.00000	1.00000	1.00000	4.4e-11	0.00629	-	-	-	-	-	-
Nudity	0.02578	0.00013	0.07268	1.00000	1.00000	0.00521	-	-	-	-	-
Racing	0.02487	1.9e-05	0.10288	1.00000	1.00000	0.00267	1.00000	-	-	-	-
RPG	1.00000	1.00000	1.00000	1.1e-10	0.00046	1.00000	0.00052	0.00054	-	-	-
Sexual Content	1.00000	1.00000	1.00000	1.0e-05	0.01791	1.00000	0.02258	0.04349	1.00000	-	-
Simulation	< 2e-16	< 2e-16	5.8e-14	1.00000	1.00000	< 2e-16	1.00000	1.00000	4.6e-13	7.8e-07	-
Sports	0.00015	1.2e-06	0.00087	1.00000	1.00000	3.5e-05	1.00000	0.95885	2.6e-06	0.00022	1.00000
Strategy	1.00000	0.00023	1.00000	0.00610	0.38867	0.15108	1.00000	1.00000	0.06953	1.00000	6.9e-05
Violent	< 2e-16	< 2e-16	< 2e-16	1.00000	1.00000	< 2e-16	1.00000	1.00000	2.3e-15	5.2e-08	1.00000
	Sports	Strategy									
Adventure	-	-									
Casual	-	-									
Free to Play	-	-									
Gore	-	-									
Indie	-	-									
Nudity	-	-									
Racing	-	-									
RPG	-	-									
Sexual Content	-	-									
Simulation	-	-									
Sports	-	-									
Strategy	0.00520	-									
Violent	1.00000	4.4e-07									

```
P value adjustment method: bonferroni
~ |
```

Interpretation:

The assumptions for both the Shapiro-Wilk test (for normality) and Levene's test (for equal variances) were violated, so the Kruskal-Wallis test was applied. From the acquired p value we learned that at least one of the genre's review_ratio was significantly different compared to the rest.

To be able to tell which exact genre differed, a pairwise Wilcoxon rank sum test was used, specifically with a correction called Bonferroni so that it could control the large amount of comparisons being made. There were two main categories: one that showed no significant difference among each other such as Action, Casual etc. and one that differed a bit more compared to the other group such as Simulation, Sports etc.. When these two groups were compared, for example Action and Simulation, a real distinction can be made. This shows that rather than individual genres being distinct, rather there are clusters and only with them can you make a separation. The only exception to this, which acted kind of like an outlier, was the Adventure genre as it stood out from both of these groups and was significantly different. Considering the results, H0 is rejected and **H1 is accepted**.

-
- Hypothesis 3 (free_to_play vs review_ratio)

```

> #Hypothesis 3
> h3_model <- t.test(review_ratio ~ free_to_play, data = steam)
> summary(h3_model)

```

	Length	Class	Mode
statistic	1	-none-	numeric
parameter	1	-none-	numeric
p.value	1	-none-	numeric
conf.int	2	-none-	numeric
estimate	2	-none-	numeric
null.value	1	-none-	numeric
stderr	1	-none-	numeric
alternative	1	-none-	character
method	1	-none-	character
data.name	1	-none-	character

```

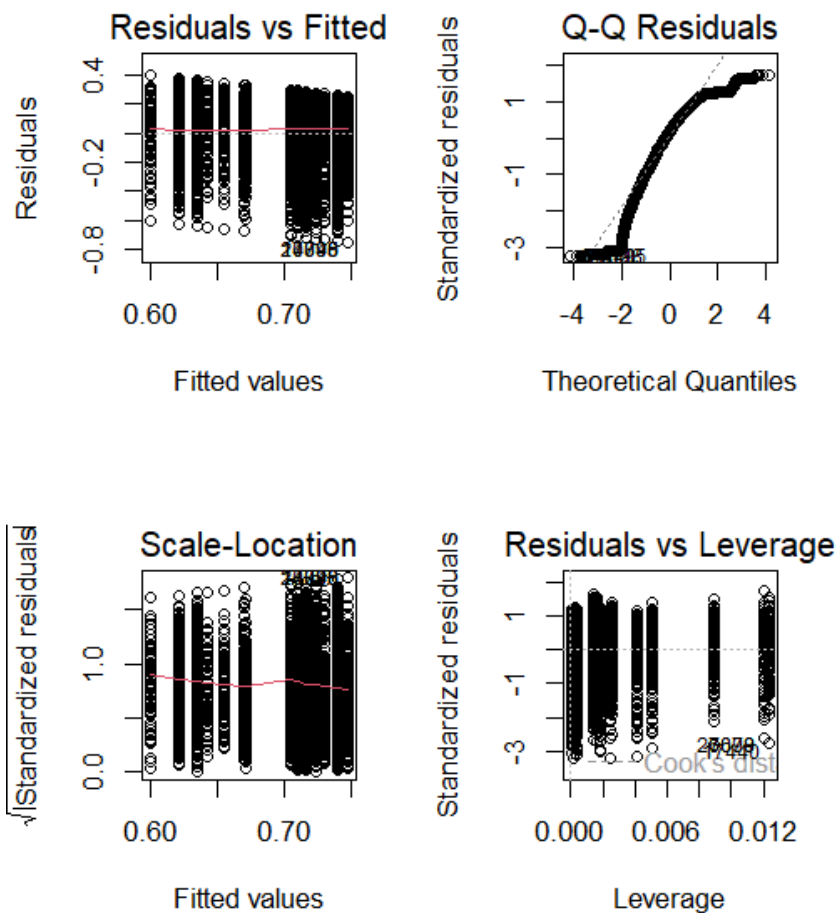
~ |

```

```

> set.seed(42)
> steam %>%
+   group_by(free_to_play) %>%
+   summarise(shapiro_p = shapiro.test(sample(review_ratio, min(5000, n())))$p.value)
# A tibble: 2 x 2
  free_to_play shapiro_p
  <lgl>         <dbl>
1 FALSE         2.00e-46
2 TRUE          7.44e-22
> par(mfrow = c(2, 2))
> plot(h2_model)
> par(mfrow = c(1, 1))
~ |

```



```
> leveneTest(review_ratio ~ free_to_play, data = steam)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value    Pr(>F)
group   1 142.46 < 2.2e-16 ***
26854
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Warning message:
In leveneTest.default(y = y, group = group, ...) : group coerced to factor.
```

```
> wilcox.test(review_ratio ~ free_to_play, data = steam)

Wilcoxon rank sum test with continuity correction

data: review_ratio by free_to_play
W = 32072139, p-value = 1.794e-05
alternative hypothesis: true location shift is not equal to 0
```

Interpretation:

Normality was violated for the Shapiro-Wilk test and the variances were unequal in Levene's test, so the Wilcoxon rank sum test was employed. The result from it was below 0.05, which shows that there is a statistically significant difference between the two variables: review_ratio and free_to_play and paid games. This is further confirmed with the boxplot that was created in the exploratory data analysis. From the results obtained, H0 is rejected and **H1 is accepted**.

- Hypothesis 4 (achievements vs average_playtime):

```
> #Hypothesis 4
> h4_model <- lm(average_playtime ~ achievements, data = steam)
> summary(h4_model)

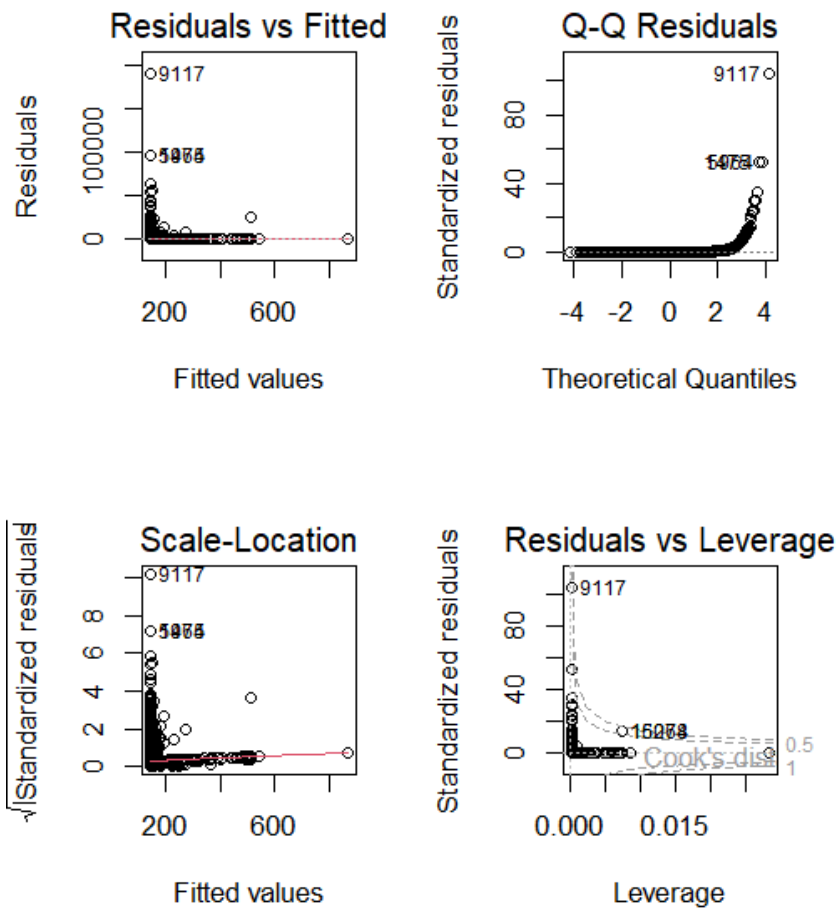
Call:
lm(formula = average_playtime ~ achievements, data = steam)

Residuals:
    Min       1Q   Median       3Q      Max
 -871    -146    -145    -145   190480

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 145.07825    11.21904   12.931  <2e-16 ***
achievements  0.07390     0.03143    2.352  0.0187 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1823 on 26854 degrees of freedom
Multiple R-squared:  0.0002059, Adjusted R-squared:  0.0001687
F-statistic: 5.531 on 1 and 26854 DF, p-value: 0.01869
```

```
par(mfrow = c(2, 2))
plot(h4_model)
par(mfrow = c(1, 1))
|
```



```
> set.seed(42)
> shapiro.test(sample(resid(h4_model), 5000))
```

Shapiro-wilk normality test

```
data:  sample(resid(h4_model), 5000)
W = 0.12621, p-value < 2.2e-16
```

```
> steam <- steam %>%
+   mutate(log_avg_playtime = log1p(average_playtime))
> h4_model_log <- lm(log_avg_playtime ~ achievements, data = steam)
> summary(h4_model_log)
```

Call:

```
lm(formula = log_avg_playtime ~ achievements, data = steam)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.222	-1.158	-1.158	-1.158	11.000

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.158e+00	1.400e-02	82.710	<2e-16 ***
achievements	6.527e-06	3.922e-05	0.166	0.868

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.276 on 26854 degrees of freedom

Multiple R-squared: 1.031e-06, Adjusted R-squared: -3.621e-05

F-statistic: 0.02769 on 1 and 26854 DF, p-value: 0.8678

```
> par(mfrow = c(2, 2))
> plot(h4_model_log)
> par(mfrow = c(1, 1))
> set.seed(42)
> shapiro.test(sample(resid(h4_model_log), 5000))
```

Shapiro-Wilk normality test

data: sample(resid(h4_model_log), 5000)

W = 0.56122, p-value < 2.2e-16

```

> summary(h4_model_log)

Call:
lm(formula = log_avg_playtime ~ achievements, data = steam)

Residuals:
    Min       1Q   Median       3Q      Max
-1.222 -1.158 -1.158 -1.158  11.000

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.158e+00  1.400e-02  82.710  <2e-16 ***
achievements  6.527e-06  3.922e-05   0.166    0.868
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.276 on 26854 degrees of freedom
Multiple R-squared:  1.031e-06, Adjusted R-squared:  -3.621e-05
F-statistic: 0.02769 on 1 and 26854 DF,  p-value: 0.8678

```

Interpretation:

As there were a lot of zero values in average_playtime, I used log1p() to sort them out and get the coefficient for achievements. It is important to note that before the transformation that the achievements coefficient did reach significance, which was assumed to be caused by the extreme skewness and hence ignored. The coefficient ($6.257e^{-06}$) turned out to be very small and not statistically significant (as the p value is 0.868, which is greater than 0.05). In addition to that, both the F statistic and R squared show that achievements are not correlated to the variance in average_playtime at all. Moving on with the Shapiro-Wilk test, there is no normality and the datapoints for average_playtime get clustered up around the 0 mark when pitted against achievements. As shown, **we fail to reject H0**.

-
- Hypothesis 5 (owners_estimate vs windows_only):

```

> #Hypothesis 5
> h5_model <- t.test(owners_estimate ~ windows_only, data = steam)
> h5_model

Welch Two Sample t-test

data: owners_estimate by windows_only
t = 4.9571, df = 9818.8, p-value = 7.276e-07
alternative hypothesis: true difference in means between group FALSE and group TRUE is not equal to 0
95 percent confidence interval:
 68608.5 158360.5
sample estimates:
mean in group FALSE mean in group TRUE
      211307.29      97822.79

```

```

> set.seed(42)
> steam %>%
+   group_by(windows_only) %>%
+   summarise(shapiro_p = shapiro.test(sample(owners_estimate, min(5000, n())))$p.value)
# A tibble: 2 × 2
  windows_only shapiro_p
  <lgl>         <dbl>
1 FALSE         8.79e-95
2 TRUE          8.17e-95
> |

```

```

par(mfrow = c(2, 2))
plot(h4_model_log)
par(mfrow = c(1, 1))

```

```

> leveneTest(owners_estimate ~ windows_only, data = steam)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value    Pr(>F)
group   1  42.568 6.947e-11 ***
      26854
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Warning message:
In leveneTest.default(y = y, group = group, ...) : group coerced to factor.

```

```

> wilcox.test(owners_estimate ~ windows_only, data = steam)

Wilcoxon rank sum test with continuity correction

data: owners_estimate by windows_only
W = 88785457, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0

```

Interpretation:

Through the Shapiro-Wilk test, normality being violated is observed (it is below 0.05 for both groups) and Levene's test for equal variances is also broken. Following this, the Wilcoxon rank-sum test was used alongside the t-test, which revealed a p value that shows a significant statistical difference between `owners_estimate` and `windows_only`. Although not as reliable, the t-test showed the exact same relationship between the two variables. Due to the obtained outcome, H_0 is rejected and instead **H_1 is accepted**.

7. Final Discussion

Hypothesis 1: There is a significant correlation between a game's price and the amount of owners. Nevertheless, because of the small effect size, it is important to keep in mind that price might not necessarily be the only factor that increases ownership.

Hypothesis 2: The mean review ratio differs significantly across genres. Based on these results, a game's genre does play a role in how positively a game is received, but it is also important to consider the limitation of this hypothesis, as only the first genre listed for a game was considered. Due to this, to be completely sure of the output of this test, a few more hypotheses need to be run with games that have multiple genres.

Hypothesis 3: There is a significant difference in review ratios between free to play and paid games. Nevertheless, it is important to know that just like Hypothesis 1, despite the difference being statistically different, the two groups' medians did not have a huge gap. Due to this, if these results were to be used somewhere, the person needs to consider that other factors may also influence the `review_ratio`.

Hypothesis 4: There is no statistically significant relationship between the number of achievements in a game and its average playtime. This means that adding more achievements does not necessarily equate to players engaging more with the game.

Hypothesis 5: There is a significant difference in the amount of owners between Windows-only games and games that are available on other platforms. This implies that although making a game not restricted to only one platform might result in higher ownership, unfortunately correlation does not necessarily mean causation and other features may play a bigger factor in more people owning the game.

8. Links

Dataset used:

Steam Store Games (Clean dataset), (Nik Davis, 2019)

<https://www.kaggle.com/datasets/nikdavis/steam-store-games/data>